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Keplerate-Type Polyoxometalates-Based Triboelectric Nanogenerator for Higher Performance via the Morphology Modulation of Blackberry Structure

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The tribe-material is the key factor affecting the performance of triboelectric nanogenerators (TENGs). Inorganic materials have higher heat resistance and stability than widely used organic materials. However, the weaker tribe-property limits the application of TENGs. Modulating surface roughness by changing particle shape and size is a simple way to increase performance for TENGs. Polyoxometalates (POMs) have unrivalled structural diversity and can self-assemble to form different nanostructures. In this study, we propose $[\{(NH_4)_{42}[Mo_{72}^{VI}Mo_{60}^{V}O_{372}(CH_3COO)_{30}(H_2O)_{72}]\cdot ca.300H_2O\cdot ca.CH_3COONH_4)\}\text{-}Mo_{132}]$ and $[\{Na_8K_{14}(VO)_2[\{(Mo^{VI})(Mo_5^{VI}O_{21})(H_2O)_3]\}_{10}\{(Mo^{VI})Mo_5^{VI}O_{21}(H_2O)_3(SO_4)\}_2\{V^{IV}O(H_2O)_{20}\}\{V^{IV}O\}_{10}(\{KSO_4\}_5)_2]\cdot 150H_2O)\}\text{-}Mo_{72}V_{30}]$ with blackberry structure which are cured and prepared into film by spin-coating technique, are used as positive tribe-materials for the first time in the field of TENGs. Keplerate-type POMs can form blackberry structures with higher dispersibility and flexibility, which can be used to control surface roughness by regulating the size of particles. The discovery proves that the particle size influences the surface roughness, which adjusts the output of TENGs. According to our findings, Mo_{132}-h-TENG generates an output voltage of 29.3 V, an output charge of 8 Nc, which is 2–3 folds higher than Mo_{132}-TENG, and a maximum power density of 6.25 $mW\cdot m^{-2}$ at 300 MΩ. Our research provides that altering the dimensional size can be an available way to raise the output of TENGs.

Introduction

As new energy and environmental problems appear, the demand for sustainable energy becomes more pressing and many energy harvesting devices have been developed.[1,2] The nanogenerator (NG) converts minute physical changes into electrical and thermal energy and is one of the best wearable technologies.[3–5] Wang developed the first triboelectric nanogenerator (TENG) in 2012 to capture the wasted mechanical energy in the environment which could be applied in wearable and self-powered sensors.[6] They offer the characteristics of being widely used, lightweight, and incredibly versatile compared with other technologies.[7–11] The tribe-material is the main factor influencing the performance of TENGs.

According to the polarities of tribe-materials, the capacity of tribe-materials to donate or lose electrons has been graded.[12,13] Organic polymers, such as nylon and rubber, are selected as positive tribe-materials because of their ease of electron loss. Still, their poor heat resistance and stability limit their use in certain circumstances.[14–16] While their tribe-polarity is tiny, the superior heat resistance and stability of inorganic materials substantially expand the application range of TENGs.[17] Researchers have extensively researched various surface treatment strategies, including surface functionalisation, chemical doping, and surface patterning, to enhance the output performance of TENGs.[18,19] Surface patterning is one of them which entails the creation of patterns on the surface to alter the morphology of the surface and raise the quantity of induced charges therein to enhance the performance of TENGs. Surface patterning can be accomplished in a variety of techniques, such as doping, etching technology, controlling particle size, etc. Among them, altering the particle size and morphology is the easiest and most generally used way to regulate the surface morphology. Finding an inorganic substance with an adjustable shape is essential.

Polyoxometalates (POMs) are a class of highly charged inorganic metal-oxygen cluster compounds, which have superior physical and chemical properties, such as low cost, straightforward synthesis and excellent redox properties.[20–28] POMs can be transferred to different nanostructures as the dopped materials which are exploited in the field of energy collection and conversion with excellent properties.[29–38] Keplerate-type POMs, with their large size and structural symmetry similar to that of fullerene (C_{60}), are expected to be an inorganic material to replace C_{60}.[39–42] Our research group used Keplerate-type POMs in the field of energy collection.[43,44] In 2016, we introduced Mo_{132} in TiO_2 as a photoanode for dye-sensitized solar energy, and the efficiency was 31 per cent higher than TiO_2-based cells. In 2021, we added Mo_{132} to the SnO_2 layer of a

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chalcogenide solar cell, which resulted in a factor of 1.03 increment in efficiency. Our earlier findings show that Keplerate-type POMs are good electron transporters and redox carriers which are anticipated to take the place of C_{60} as inorganic fullerenes in TENGs. However, the tiny triboelectric polarity and readily aggregatable properties of POMs decrease the rate of electron transmission, and the way of increasing the TENG performance is eagerly explored. Liu and Prof. Müller made a lot of observations about the counterion binding properties of ($\{(NH_4)_{42}[Mo_{72}^{VI}Mo_{60}^{V}O_{372}(CH_3COO)_{30}(H_2O)_{72}]\cdot ca.300H_2O\cdot ca.CH_3COONH_4\}$-$Mo_{132}$) and ($\{Na_8K_{14}(VO)_2[\{(Mo^{VI})(Mo_5^{VI}O_{21})(H_2O)_3\}]_{10}\{(Mo^{VI})Mo5^{VI}O_{21}(H_2O)_3(SO_4)\}_2\{V^{IV}O(H_2O)_{20}\}\{V^{IV}O\}_{10}(\{KSO_4\}_5)_2]\cdot 150H_2O\}$-$Mo_{72}V_{30}$). In the mixed polar solvent of water and acetone, Mo_{132} and $Mo_{72}V_{30}$ can bind the surrounding counterions to form larger blackberry structure (Mo_{132}-h and $Mo_{72}V_{30}$-h), which are thermodynamically stable and more flexible.[21,22] As a result, the blackberry-structured POMs as positive tribe-materials have several excellent advantages. Firstly, different nanostructures can reduce the agglomeration of POM particles and increase the electron transport rate to improve the performance of TENGs; Secondly, the different nanostructures controllably regulate the surface morphology of materials to increase the surface tribe-charge; Thirdly, the nanostructures of POMs possess higher flexibility and stability, which can be applied to the wearable field under harsh conditions.

In this study, we have assembled a TENG with polyethene terephthalate (PET) as the negative tribe-layer and Mo_{132}-h and $Mo_{72}V_{30}$-h as positive tribe-layers for the first time, which are cured and formed as films via the spin-coating technique. The single scattered ion is transformed into the larger size blackberry structure making use of the counterion effect. Our finding demonstrates that the surface roughness changes as the size of Mo_{132} and $Mo_{72}V_{30}$ blackberry structures increase, which leads to an improvement in the performance of TENGs. Mo_{132}-h-TENG can achieve an open-circuit voltage of 29.3 V, short current of 0.75 mA $\cdot m^{-2}$, and transfer charge of 8 nC respectively, which continue to run steadily and continuously for more than 2500 pushes. The maximum power amounts to 6.25 mW$\cdot m^{-2}$ at 300 MΩ external resistance. Furthermore, our research indicates that altering the size is a practical way to enhance the efficiency of TENGs.

Results and Discussion

Characterizations of Blackberry Structure and Film

We assembled Mo_{132} and PET using four sponges that were spaced apart and attached copper tapes to the exterior circuit. The process of assembling is specifically depicted (see Figure 1a).

The precursors for the blackberry structure are Mo_{132} and $Mo_{72}V_{30}$, which have been successfully prepared by the FT-IR spectra (see supporting information Figure S1). The distinctive peaks of $Mo_{72}V_{30}$ are loaded at 1613 cm^{-1}, 1195 cm^{-1}, 1119 cm^{-1},1052 cm^{-1},957 cm^{-1},771 cm^{-1},621 cm^{-1}, and 553 cm^{-1}, while the typical peaks of Mo_{132} are present at 1615 cm^{-1}, 1523 cm^{-1}, 1409 cm^{-1}, 936 cm^{-1}, 846 cm^{-1},

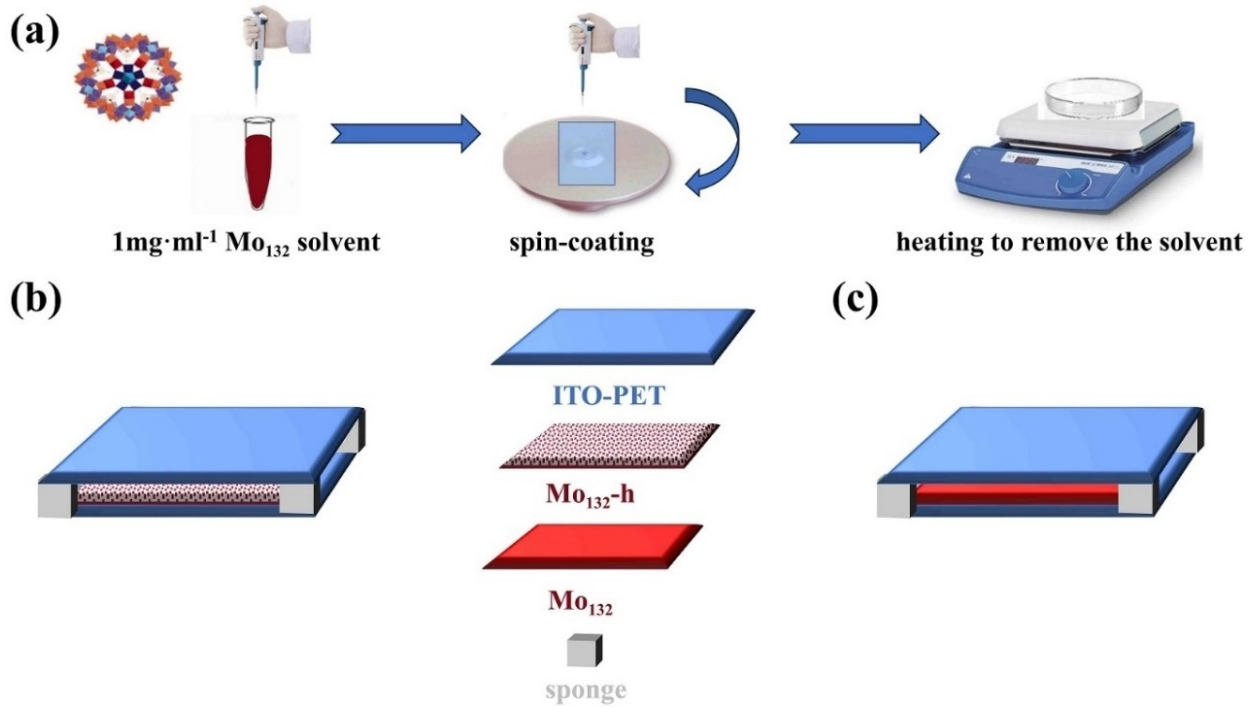


Figure 1. The assembling process and schematic diagram of TENG. (a) Assembling process of Mo_{132}-h-TENG. (b) Schematic diagram of Mo_{132}-h-TENG after surface patterning. (c) The structure diagrams of Mo_{132}-TENG and Mo_{132}-h-TENG

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780 cm^{-1},711 cm^{-1}, and 621 cm^{-1}.[45,46] Mo_{132} and $Mo_{72}V_{30}$ are both big oxo-metallate clusters with environmental, easy synthesis excellences, which can act as triboelectric positive materials carrying many positive charges.[47] The structures of original materials can be transferred from discrete macroion to big-size blackberry in the mixture of acetone and water. Inside, the motive of the blackberry formation is not Van der Waals and hydrophobic intervention, nevertheless, it is the associative effect between macroion and counter ion in the polar solvent that plays a key role in the process in which the discrete macroion can transfer to blackberry. In the meantime, the formation of blackberries is also influenced by the volume ratio of acetone, and the volume fraction of acetone loads between 3% and 75% for Mo_{132}-h generation; the volume fraction of acetone is at the following 75% for $Mo_{72}V_{30}$-h formation.

The morphology and element distribution of the blackberry structure were characterized by TEM, which identifies the blackberry structure was successfully prepared (see Figures 2a–c and supporting information Figures S2-4). For the precursor resolvents of Mo_{132}, our research group chose 5%, 10%, 30%, and 40% acetone solvents, and 5%, 10%, 30% and 65% acetone solvents as the precursor resolvents for $Mo_{72}V_{30}$. The sizes of Mo_{132}-h and $Mo_{72}V_{30}$-h are determined to be around 45 nm and 20 nm, respectively. (see Figures 2a-b and supporting information Figure S3). Element mapping was used to prove that the Mo and N elements are evenly distributed at the blackberry structure (see Figure 2c). $Mo_{72}V_{30}$-h were created in succession illustrated by the mapping of $Mo_{72}V_{30}$–h (see supporting information Figures S3–4), which also shows that the V, Mo and N elements are equally dispersed at the structure.

Besides, SEM was utilized to characterize the surface morphology and element distribution of the film (see Figures 2d–f and supporting information Figure S5). The film of Mo_{132}-h, which lacks needle-shaped nicks and cracks with stronger completeness and denseness than the film for Mo_{132}, is coated in noticeable particles. The element mapping of Mo_{132}-h indicates that Mo and N elements are equally dispersed in the film, which further proves the blackberry structure was successfully prepared as a film and kept stabilized in the solid state. The film for $Mo_{72}V_{30}$-h has higher dispersibility than the film of $Mo_{72}V_{30}$. The element mapping proves V, Mo and N element uniformly decentralizes in the film. We also tested the cross-sectional thickness of the thin film because of different particle sizes. The thickness of ITO is approximately 100 nm, and the film adheres to the surface of ITO (see Supporting Information Figure S6). The thickness of Mo_{132}-h film is about 10 nm, which is higher than that of Mo_{132} film at 5 nm and the thickness of $Mo_{72}V_{30}$ film is approximately 12 nm, which is lower than that of $Mo_{72}V_{30}$-h film at 20 nm. $Mo_{72}V_{30}$ has a higher concentration, therefore its thickness is higher than Mo_{132}. This further indicates that there are differences in particle size.

The Assembling and Electrical Output for TENGs

Our research group evaluated the electrical capabilities of Mo_{132}-TENG compared to the Mo_{132}-h-TENG. Firstly, PET is utilized as the triboelectric negativity material, and the copper tape connects the device with the external circuit, which clings to the ITO surface of the ITO-PET. Next, Mo_{132} (Mo_{132}-h) functions as the triboelectric positive material, and the copper tape is connected to the ITO surface which is covered by the Kapton tap. Finally, two triboelectric layers are separated by four sponges, and the copper tape connects all of the electrical components in the circuit.

The Mo_{132}-TENG outputs are displayed (see Figures 3c–e). The values of voltage, short current, and charge of Mo_{132}-TENG

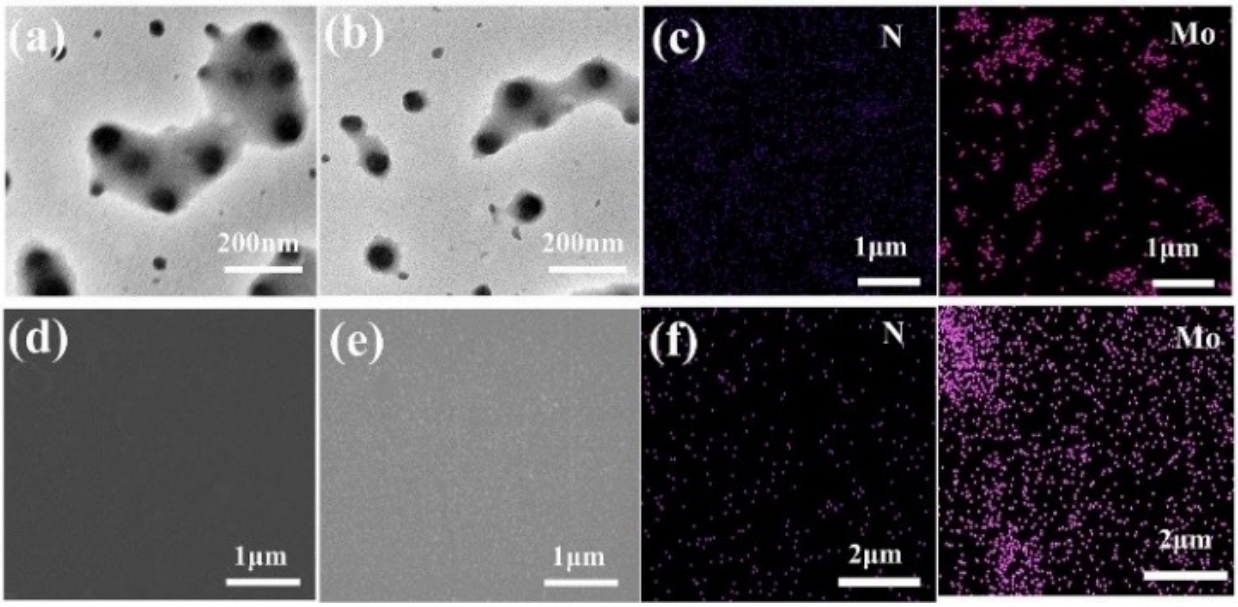


Figure 2. The morphology and element distribution of Mo_{132}-h and film. (a–b) TEM images, (c) element mapping of Mo_{132}-h. (d–e) SEM images of films for Mo_{132} and Mo_{132}-h, and (f) element mapping of film for the Mo_{132}-h

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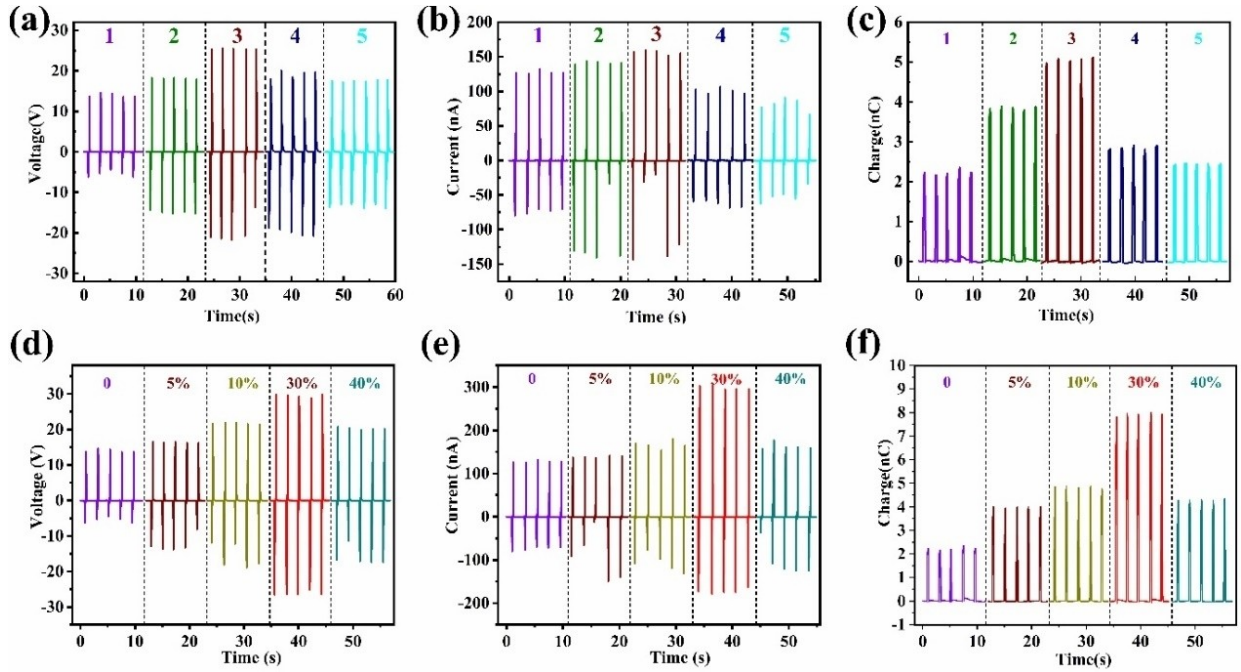


Figure 3. The outputs of Mo_{132}-TENG and Mo_{132}-h-TENG. (a) Voltage, (b) current, and (c) charge curves of Mo_{132}-TENG at different concentrations of Mo_{132}. (d) Voltage, (e) current, (f) charge curves of Mo_{132}-h-TENG at different ratios of acetone and water.

are each 13.7 V, 128.9 nA and 2.2 nC, respectively, while the molecular weight of Mo_{132} is 1 mg·ml^{-1}. Additionally, as the concentration of Mo_{132} increased, the voltage, current and charge increased as well, reaching their maximum values at a concentration of 3 mg·ml^{-1} and were all at their respective maximum values of 25.7 V, 160 nA and 5.1 nC. A decrease in output is caused by the higher concentration. The voltage, current and charge, which were previously 12.3 V, 80 nA and 2.5 nC, respectively, at the concentration of 5 mg·ml^{-1} due to the Mo_{132} discrete macroion aggregating to impede the ability of electronic transmission. The performance changes of $Mo_{72}V_{30}$-TENG are consistent with the pattern of $Mo1_{32}$-TENG. The performances of the $Mo_{72}V_{30}$-TENG are displayed and the best performance is attained at 3 mg·ml^{-1}, with the voltage, current and charge values corresponding to 24.5 V, 156 nA and 4.2 nC (see supporting information Figure S7).

Comparing the effect of particle size on TENG performance, we tested the output performances of Mo_{132}-h-TENG (see Figures 3d–f). When the finger pressed the device, the values of voltage, current, and charge can all be as high as 29.8 V, 0.75 mA ·m^{-2}, and 8 nC respectively and these values are independently multiplied by 2.17, 2.33 and 3.63 times as the volume proportion of ace-tone is 30%. The size of Mo_{132}-h has a significant impact on the performance of TENG. The performance of $Mo_{72}V_{30}$-h-TENG has enhanced as compared to the performance of $Mo_{72}V_{30}$-TENG (see supporting information Figure S8). The voltage increases to 24.5 V, the current improves to 0.64 mA ·m^{-2}, and the charge increases to 7.5 nC. The explanation for the increment in output is consistent with the Mo_{132}-h-TENG and further demonstrates that the size affects performance for TENG.

We further investigated the strength and stability of Mo_{132}-h and $Mo_{72}V_{30}$-h-TENG in further detail. The voltage, current curve, and power are shown as various loads connected to the external circuit (see Figures 4a–c and supporting information Figures S8–9). Our team connected the load to the circuit and gradually altered the value of the load. The finding demonstrates that the current continuously reduces and the voltage continuously rises with the various loads connected in the external circuit, which is consistent with the regularity of TENGs. When the load is 300 MΩ for the Mo_{132}-h-TENG, the instant power reaches its maximum magnitude of 6.25 mW·m^{-2}. In Figure 4c, the value of voltage for Mo_{132}-h-TENG maintains 30 V even after being pressed more than 2500 times, demonstrating the high level of persistence for the device. The voltage, current curve, power and stability of $Mo_{72}V_{30}$-h-TENG are shown (see supporting information Figures S9-10). These findings showed that the maximum power for $Mo_{72}V_{30}$-h-TENG reaches the value of 3.75 mW·m^{-2} and the device has good stability. At the same time, we tested the long-term stability and heat resistance of the device. After 144 hours, the performance of the device remains above 85%, which proves that the device has high stability (see Supporting Information Figure S11). As the temperature increases, the performance of the device does not show a significant decrease. The output voltage of Mo_{132}-h-TENG is 25 V, and the output performance of $Mo_{72}V_{30}$-h-TENG is 22 V at 100 °C, which proves that the device has high heat resistance (see Supporting Information Figure S12

Finally, we tested the electronegativity of different materials. The current of Mo_{132}-Mo_{132}-h is higher than that of Mo_{132}-Mo_{132}, indicating that they have different electronegativity, and the pattern of $Mo_{72}V_{30}$ and $Mo_{72}V_{30}$-h is also consistent with it

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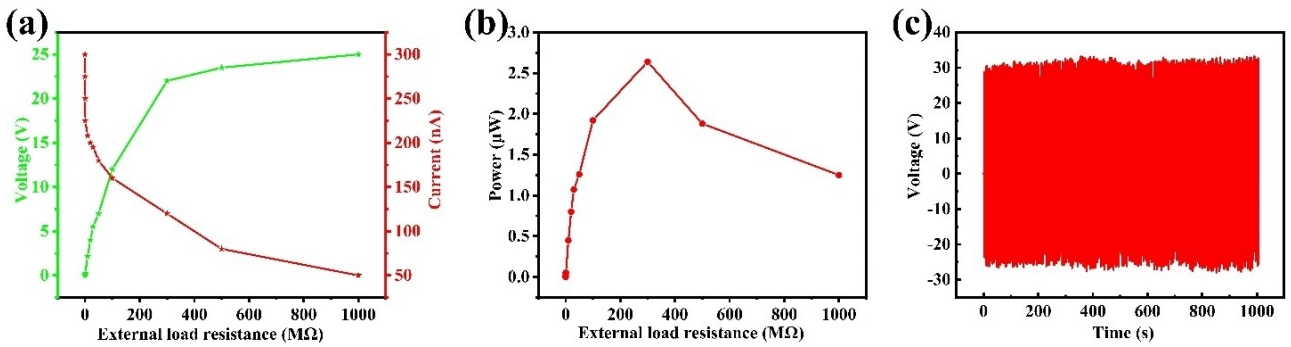


Figure 4. The power and stability curves of Mo_{132}-TENG and Mo_{132}-h-TENG. (a)Voltage and current curves, and (b) power of Mo_{132}-h-TENG at different resistances. (c) The stability at many presses.

The Mechanism of TENG

In this research, we used PET as the triboelectric negative material, which is located in the negative triboelectric series and only below the polytetrafluoroethylene (PTFE), polydimethylsiloxane (PDMS), Kapton and fluorinated ethylene propylene (FEP) in triboelectric polarity. Among the triboelectric materials, it has a higher electronic attraction ability and larger negative triboelectric polarity. Mo_{132} and $Mo_{72}V_{30}$ are big polyoxometalates with a 1 nm size, which have positive charge attraction ability and are used as triboelectric positive materials.

The processes of triboelectric electrification and electrostatic induction are presented between Mo_{132} and PET.[13] PET and Mo_{132} are separated and have no current generation in the external circuit as illustrated in Figure 5a. In Figure 5ai, PET and Mo_{132} approach each other and crimp, the PET gets negative charges as the larger electron attraction ability and the Mo_{132} induces positive charges with the smaller triboelectric polar. At this time, two triboelectric layers stand in the charge equilibrium state, which is known as triboelectric electrification. In Figure 5aii, PET and Mo_{132} gradually part, the back electrode of PET induces positive charges, and the back electrode of Mo_{132} obtains negative charges as a result of electrostatic induction. At this time, the electron inflows from the back electrode of Mo_{132} to the other side electrode of PET, and a current is formed in the external circuit. As two triboelectric layers are separated to maximum distance, the triboelectric layers reach the balanced status without external current in Figure 5aiii. In Figure 5aiv, two triboelectric layers gradually approach one other and a current is produced. Whilst the distance is best between PET and Mo_{132}, the maximum current is formed and the distance comes to the maximum extent, the charge attains the equilibrium in two triboelectric layers.

AFM was united to clarify the cause for the performance improvement of the Mo_{132}-h-TENG compared with Mo_{132}-TENG. The surface state is displayed which is spin-coated by Mo_{132}-h. The films of Mo_{132} and Mo_{132}-h are composed of many particles, only Mo_{132}-h film has higher surface roughness which is 3.179 nm and 2.27 times than 1.399 nm. 3D AFM further identified the Mo_{132}-h film has more significant roughness than the Mo_{132} membrane (see Figure 6b and Figure 6d). The size of the TENG performance is determined by the triboelectric charge density, dielectric properties and assembled structure by the knowledge of Maxwell's equation.[48] Inside, the triboelectric charge density is the minor influencing factor which is affected by the surface roughness.[49,50] The film of Mo_{132}-h has a bigger surface roughness which will generate more surface triboelectric charges to improve the performance of the TENG. The

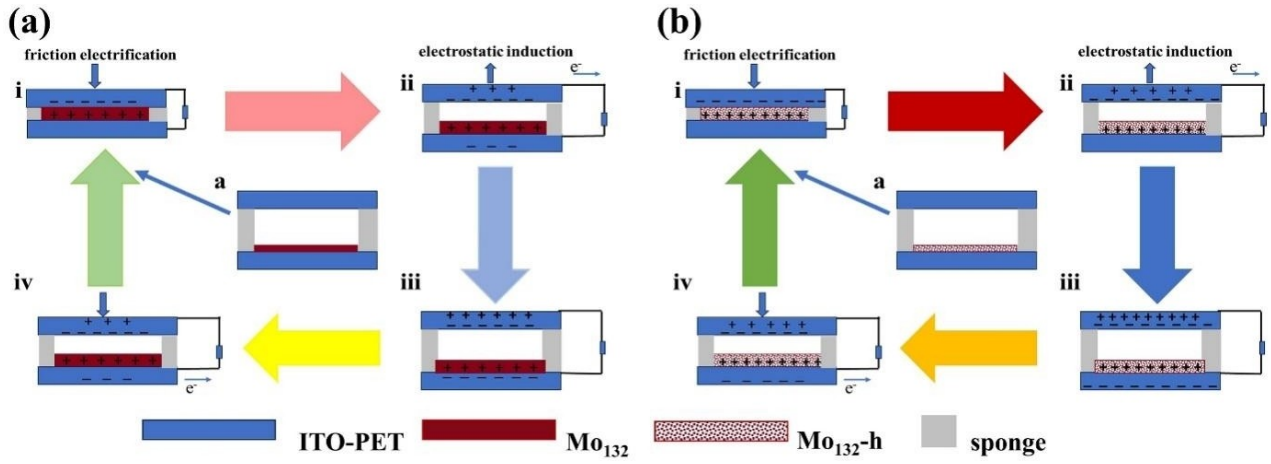


Figure 5. (a) The working mechanism of Mo_{132}-TENG. (b) The effect for Mo_{132}-h-TENG after particle morphology modification.

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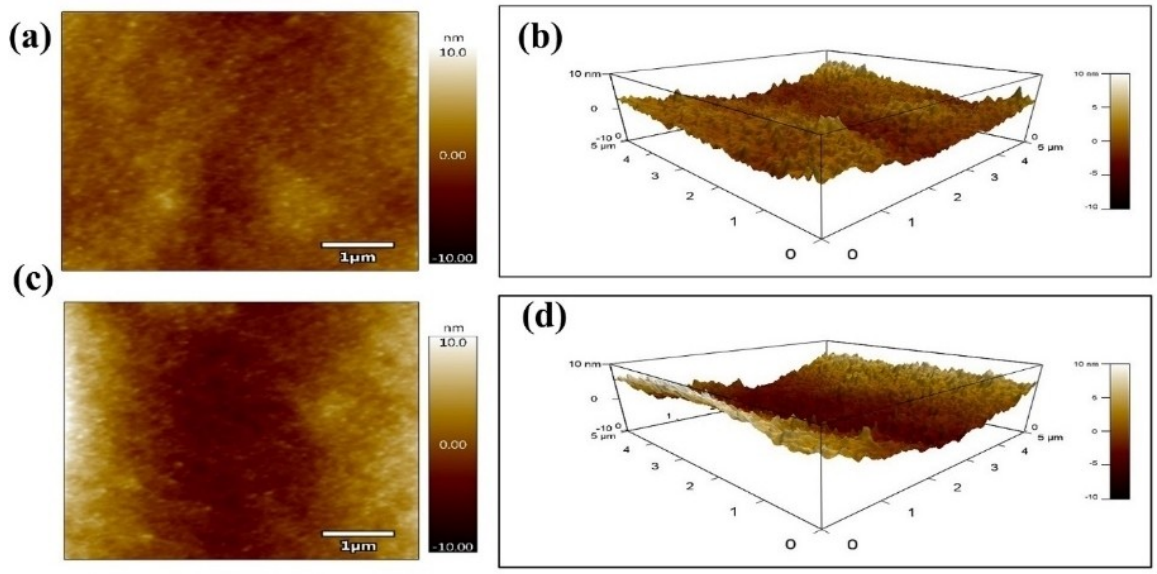


Figure 6. (a) Flat AFM images and (b) 3D AFM for the film of Mo_{132}. (c) Flat AFM images and (d) 3D AFM for the film of Mo_{132}-h.

processes of triboelectric electrification and electrostatic induction are shown among Mo_{132}-h and PET (see Figure 6d). The mechanism of $Mo_{72}V_{30}$ and $Mo_{72}V_{30}$-h-based TENGs are the same as the Mo132 and the $Mo_{72}V_{30}$ film which is respectively 2.404 nm and 1.382 nm (see supporting information Figure S13 and Figure 7).

To further explain the reasons for the performance differences, we conducted KPFM testing. The material has a higher surface potential, which has a stronger ability to gain and lose electrons. The surface potential of Mo_{132}-h film is −2.72 V, which is higher than that of Mo_{132} film at −3.02 V. This indicates that Mo_{132}-h film has stronger electron harvesting ability which is consistent with the pattern of performance.

The Tests of Mo_{132}-h-TENG in the Application

The digital images of Mo_{132}-h and Mo_{132}-h-based TENGs are presented (see Figure 8a and supporting information S15). The word TENG can be seen lightly throughout the film and device which is a proven application in successful wearable areas. The flexibility of the film and device are tested (see Figure 8b). The Mo_{132}-h film and TENG have good suppleness, which further proves the Mo_{132}-h-TENG can be used in wearable areas. Our research group fixed the device on our elbow, hand, thumb finger and foot and tested the output in different body parts. The device can emerge in different voltage values and shapes due to the distinct sport states, which can separately generate 15 V, 6 V, 20 V and 40 V. Therefore, the Mo_{132}-h-TENG can be used to judge the sport form.

Conclusions

In summary, the blackberry structure was successfully prepared which has a bigger size by an easy strategy of counter ion association affection. We adjusted the size of the particle to regulate the surface roughness of the film which was prepared by the spin-coating method. The performance of the device which is assembled with PET and Mo_{132}-h, is improved as a

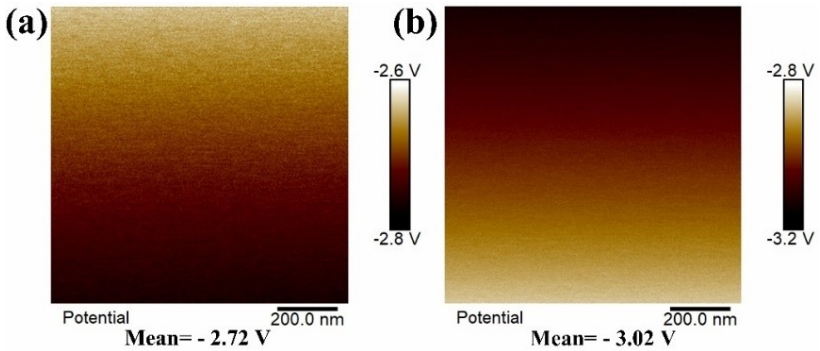


Figure 7. KPFM morphologies of (a) Mo_{132}-h, (b) Mo_{132}.

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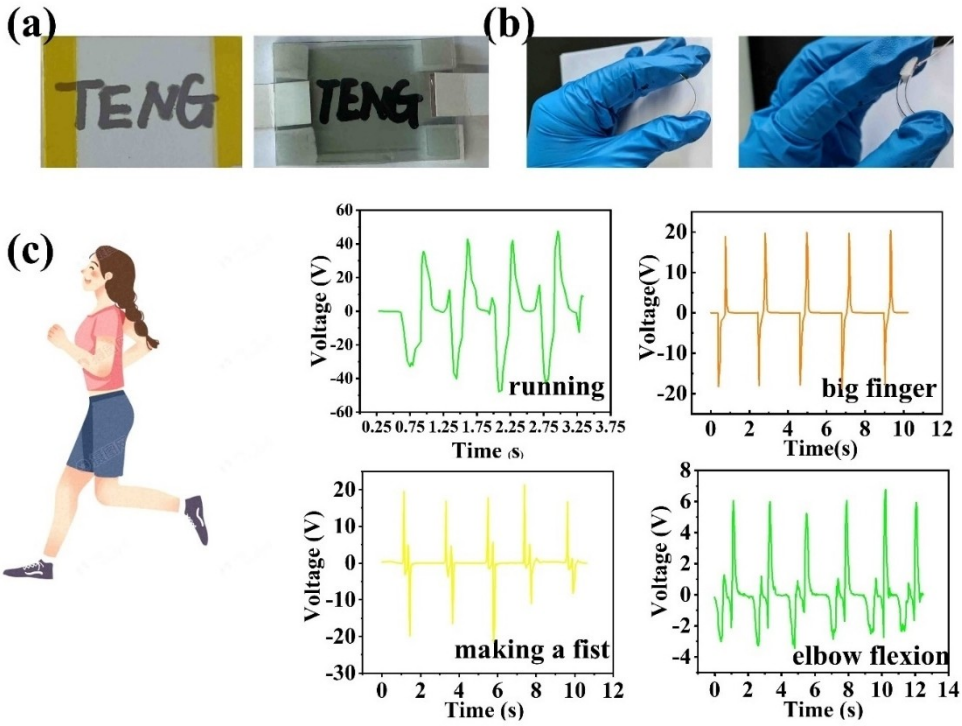


Figure 8. (a) Digital image of Mo_{132}-h film and Mo_{132}-h-TENG. (b) The flexibility for the film and TENG of Mo_{132}-h. (c) Wearable applications of Mo_{132}-h-TENG.

result of the increment for surface roughness, which was characterized by 6517B and 6500. From these results, the performance of the Mo_{132}-h-TENG is optimum as the acetone ratio is 30%. The value of voltage is 29.8 V, the value of current is 300.3 nA, the value of charge is 8 nC, and the peak power amounts to the value of 2.6 μW at 300 MΩ. These results further validate that the Mo_{132}-h-TENG has great prospects in wearable areas. On the whole, our research achievements contribute to seeking more environmentally and efficient positive tribo-electric materials and offer a sustainable view on performance improvement.

Experiment Section

The Reagents and Materials

All the reagents and materials that were not further purified were used in Aladdin. The precursors (Mo_{132} and $Mo_{72}V_{30}$) were made per the suggested procedures in the literature[39,40] Mo_{132}-h and $Mo_{72}V_{30}$-h, the target materials, were synthesized by group Liu.[21,22,39] Acetone and deionized water were used as the solvents for the composition of Mo_{132}-h and $Mo_{72}V_{30}$-h. The combined solvent has acetone contents of 5%, 10%, 30%, and 40%, and 1 mg of Mo_{132} was added individually to mixtures of different ratios. Mo_{132}-h was produced in the solution after each centrifuge tube had been individually sealed to keep out air and placed in the oven for a few days. The preparation technique for $Mo_{72}V_{30}$-h is the same as for Mo_{132}-h; the only differences were the weight ratio of $Mo_{72}V_{30}$, which was 5 mg·ml^{-1}, and the percentage of acetone, which was, respectively, 5, 10, 30 and 65.

The Assembly of TENGs

Using Mo_{132} as an example, 1 mg·ml^{-1} of the solution was spin-coated on the Indium Tin Oxides (ITO) side of an ITO-PET surface and left there for 30 minutes while the oven was at 80 °C to remove the solvent, which served as the positive triboelectric layer with a 4 cm^2 surface area. Additionally, the polyethylene terephthalate (PET) layer was utilized as the negative triboelec-tric layer. The final step involved assembling Mo_{132} and PET using four sponges that were spaced apart and attaching copper tapes to the exterior circuit. The process of assembling is specifically depicted (see Figure 1a), the other TENGs also follow the same procedure. The final schematic representations of Mo_{132}-TENG and Mo_{132}-h-TENG are shown (see Figures 1b–c).

The Characterizations of Materials

The compositions of Mo_{132} and $Mo_{72}V_{30}$ were examined using an Alpha Centauri FT/IR spectrophotometer (FT-IR). The morphology and element content of Mo_{132}-h and $Mo_{72}V_{30}$-h were determined using an energy-dispersive spectrometer (EDS) and high-resolution transmission electron microscopy (HRTEM) (JEOL-2100 F). The surface topography and element distribution of the film were obtained using energy-dispersive X-ray spectroscopy (EDX) and scanning electron microscope (SEM Hitachi S-8010), respectively. The enhancement of blackberry-based TENG was explained by using the atomic force microscope (AFM) to characterize the surface roughness.

The Performance Tests

Three performance metrics (V, I and Q) were measured using an electrometer (Keithley 6517B) and a digital multimeter (DMM 6500).

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The power was measured by combining different resistances with the external circuit.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: blackberry structure · polyoxometalates · polyanion · polymetallic oxygen cluster · triboelectric nanogenerator

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